GeoNatureAgent Benchmark: Benchmarking LLM Agents for Environmental Geospatial Analysis Across Frontier and Open-Weight Foundation Models [Benchmark]

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Abstract

Environmental scientists spend disproportionate effort on data wrangling rather than analysis. New AI agents can be a helpful tool, but their use still requires validation. Furthermore, no benchmark exists to evaluate AI agents that automate environmental geospatial workflows through structured tool calling against real APIs. We introduce the **GeoNatureAgent Benchmark**, the first benchmark for environmental analysis agents that operate via structured tool calls to a production-style geospatial API. The benchmark comprises 93 tasks across 18 categories, covering municipality-level analysis, multi-turn conversation, spatial reasoning, cross-indicator synthesis, error handling and recovery, ranking, comparison, multilingual understanding, habitat analysis, deep-dive profiling, temporal change, and task rejection. Tasks are evaluated against an open, self-hostable geospatial API that serves three environmental indicators across Spain and Portugal via twelve domain-specific tools. We evaluate seven large language models (Claude Sonnet 4, DeepSeek V3.2, GLM-5, Gemini 2.5 Pro, Qwen3-235B, GPT-OSS-120B, and Llama 4 Scout) across Vertex AI and Anthropic platforms under three temperature-1.0 seeds per model, reporting capability and per-case cost as orthogonal axes. Results manifest that: (1) Claude Sonnet 4 achieves the highest capability at 60.8% ± 0.8%, followed closely by DeepSeek V3.2 at 56.3% ± 3.1%, while no other model exceeds 51%; (2) the cost-accuracy Pareto frontier is occupied entirely by open-weight models (Llama 4 Scout → Qwen3-235B → DeepSeek V3.2), with DeepSeek V3.2 offering 93% of Claude's capability at 11× lower cost($0.011/case); (3) comparison tasks remain universally unsolved (0% across all models for close-value comparisons), exposing systematic reasoning limitations; and (4) structured tool calling against a real API provides a more discriminative measure of real-world agent capability, with mean accuracies 25–35 percentage points below those reported on general-purpose GIS benchmarks. We further demonstrate geographic and domain extensibility by integrating BigEarthNet V2 land-cover data for Portugal alongside Spanish CO_2 suitability and gully erosion indicators. GeoNatureAgent Benchmark, the evaluation harness, and the self-hostable API are publicly available[1].

CCS Concepts

• **Computing methodologies → Neural networks**; *Intelligent agents*; *Multi-agent systems*; *Information extraction*; • **General and reference → Evaluation**; • **Information systems → Geographic information systems**; • **Applied computing →** *Earth and atmospheric sciences.*

Keywords

Geospatial AI, Benchmark, LLM Agents, Tool Calling, Environmental Analysis

1 Introduction

Environmental monitoring at scale requires multi-temporal geospatial analysis, including the detection of land cover change, the assessment of erosion risk, or the computation of vegetation indices; furthermore, the generation of statistical summaries for regulatory compliance. Practitioners routinely spend the majority of their effort on data wrangling—discovering datasets, reprojecting coordinate systems, and debugging code against remote sensing APIs [13, 28]. This expertise barrier limits the adoption of geospatial methods by domain scientists who understand environmental systems but lack GIS programming skills.

Recent advances in large language models (LLMs) have led to the development of AI agent systems that translate natural language into executable geospatial operations. Architectures range from LLMs with tool pools [1, 28] to fine-tuned specialists [26, 27] to multi-agent systems achieving 85–97% accuracy [12, 14]. Major industry players are investing heavily: Google's Earth AI integrates Gemini with AlphaEarth Foundations [6], the Planet-Anthropic partnership applies Claude to daily satellite imagery [18], and

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[1] https://github.com/gabrielireland/GeoNatureAgent_Benchmark

NASA/IBM's Prithvi-EO-2.0 provides pre-trained vision transformers [22].

However, a critical evaluation gap persists. Public benchmarks for LLM agents in geospatial settings remain scarce, and the few that exist target general GIS tasks (GeoBenchX [10], ThinkGeo [19]) rather than environmental science workflows. Environmental knowledge benchmarks (EnviroExam [7]) evaluate knowledge, not agent behavior. Code-generation benchmarks (UnivEARTH [8]) do not reflect the structured API interaction that production systems employ—58% of LLM-generated Earth Engine code fails to execute. At the foundation model level, GEO-Bench [11] and GEO-Bench-2 [20] evaluate pixel-level vision models, not the agent-level tool orchestration that determines real-world usability.

This work fills this gap through four main contributions:

(1) **GeoNatureAgent Benchmark**—the first benchmark for environmental analysis agents using structured tool calling against a production-style geospatial API, comprising 93 tasks across 18 categories with twelve domain-specific tools, evaluated by eight mechanistic checks per case (no LLM-as-judge).
(2) **Multi-seed cross-platform evaluation**—a systematic comparison of seven LLMs spanning frontier closed-source models (Claude Sonnet 4, Gemini 2.5 Pro) and frontier open-weight families (DeepSeek V3.2, GLM-5, Qwen3-235B, GPT-OSS-120B, Llama 4 Scout), each evaluated under three temperature-1.0 seeds on identical infrastructure with mean-±-standard-deviation reporting; capability and per-case cost are reported as orthogonal axes rather than collapsed into a single gated score.
(3) **Cost-efficiency Pareto analysis**—quantitative cost-accuracy mapping revealing that the cost-accuracy Pareto frontier is occupied entirely by open-weight models, with DeepSeek V3.2 offering 93% of Claude's capability at 11× lower cost.
(4) **Reproducible MLOps pipeline and open API**—a fully automated Cloud Build pipeline that produces conversation-trace-level results in object storage, plus an Apache-2.0 self-hostable FastAPI service so the entire evaluation can be reproduced without any private endpoint.

The remainder of this paper is organized as follows: Section 2 reviews related work in geospatial AI agents and benchmarks. Section 3 describes the GeoNatureAgent Benchmark design, the ReAct-style agent architecture, and the evaluation protocol. Section 4 presents the experimental results, including the cost-accuracy Pareto analysis of the seven evaluated models. Section 5 provides a discussion comparing the findings with prior benchmarks and exploring domain extensibility. Finally, Section 6 summarizes the conclusions and suggests future work.

2 Related Work

2.1 Geospatial AI Agent Architectures

The literature reveals four paradigms. The **LLM + Tool Pool** approach pairs general-purpose LLMs with callable tools: GeoGPT [28] uses GPT-3.5-turbo with LangChain, LLM-Geo [13] uses GPT-4 for DAG-based decomposition, and GIS Copilot [1] integrates LLMs into QGIS. **Fine-tuned specialists** sacrifice generality for expertise: GTChain [26] fine-tuned LLaMA-2-7B, achieving 32.5% higher accuracy than GPT-4; EnvGPT [27] fine-tuned an 8B model on 100M environmental tokens, rivaling GPT-4o-mini. **Multi-agent systems** decompose tasks: GeoJSON Agents [14] achieved 97% with GPT-4o Planner-Worker architecture vs. 49% with a single-agent; GeoLLM-Squad [12] gained 17% from specialized sub-agents. **Cost-efficient hybrids**: Geo-OLM [21] enables sub-7B models to perform within 10% of GPT-4o at 100× lower cost.

2.2 Benchmarks and Evaluation

Agent tool-use is evaluated by GeoBenchX [10], ThinkGeo [19] (486 remote sensing tasks), and UnivEARTH [8] (only 33% accuracy on Earth Engine code generation and a 58% code failure rate). GeoBenchX is the closest comparable work to GeoNatureAgent Benchmark: 202 tasks across five categories (merge-visualize, process-merge-visualize, spatial operations, heatmaps/contour lines, and control questions), evaluated via 24 general-purpose GIS tools (data loading, filtering, spatial joins, and choropleth rendering) with an LLM-as-judge scoring protocol (3-point scale, three independent judges achieving 88–96% agreement with human annotations). Top performers are o4-mini and Claude 3.5 Sonnet. Crucially, GeoBenchX tools are low-level primitives operating on local files—`load_data`, `filter_categorical`, `create_buffer`, `make_choropleth_map`—whereas GeoNatureAgent Benchmark tools are high-level domain operations against a production cloud API.

To bridge the gap between general-purpose GIS benchmarks and domain-specific environmental analysis, we integrate four canonical GIS operations from GeoBenchX [10] into GeoNatureAgent Benchmark's toolset: proximity buffering (`create_buffer`), feature selection by spatial relationship (`select_features_by_spatial_relationship`), centroid extraction (`get_centroids`), and explicit task rejection (`reject_task`) to prevent hallucination on out-of-scope queries. These tools are adapted from GeoBenchX's local-file paradigm to operate against GeoNatureAgent Benchmark's cloud API and administrative boundary system. We additionally introduce 10 benchmark cases inspired by GeoBenchX's task patterns—buffer-based proximity analysis, spatial joins, and centroid calculations—applied to Spanish environmental indicators. This creates a direct comparison pathway: models achieving 58% on GeoBenchX's general GIS tasks can be evaluated on equivalent spatial operations within our domain-specific context, isolating the performance impact of domain complexity from tool-use capability.

Map reasoning: MapEval [5] (no model exceeds 67%), MapQA [2] (humans 91%, models <50%). Spatial reasoning: GPSBench [23], SpatiaLab [24] (humans 88%, best model 55%). Foundation models: GEO-Bench [11] and GEO-Bench-2 [20] evaluate *pixel-level* classification/segmentation across 19 datasets, and PANGAEA [15] finds that foundation models do not consistently outperform supervised baselines.

Key distinction: GEO-Bench evaluates pixel-level model accuracy (*can this model classify a satellite tile?*); GeoNatureAgent Benchmark evaluates agent-level tool orchestration (*can this agent reason about which tools to call?*). Both are complementary.

2.3 Environmental Science AI

EnvGPT [27] shows that focused fine-tuning on environmental tokens rivals GPT-4o-mini. Google's Earth AI [6] achieves 82%

on Q&A (vs 50% baseline Gemini). Pan et al. [17] demonstrate MCP-grounded air quality monitoring with a factual accuracy of 4.78/5.0. Spain's RD 214/2025 [9] mandates carbon footprint reporting for ~4,000 organizations, creating demand for automated multi-temporal geospatial analysis.

2.4 The Missing Benchmark

No existing benchmark evaluates environmental analysis agents performing structured tool calling against a real geospatial API with multi-turn conversation, multilingual inputs, and cross-indicator synthesis. GeoNatureAgent Benchmark fills this gap.

3 GeoNatureAgent Benchmark

3.1 Benchmark Design

GeoNatureAgent Benchmark comprises 93 tasks organized into 18 categories (see Table 1), spanning easy (19%), medium (45%), and hard (36%) difficulty levels.

Table 1: GeoNatureAgent Benchmark v5 task categories (93 tasks, 18 categories).

Category	Tasks	Tests
Tool selection	21	Correct tool choice for varied queries
Cross-indicator	8	CO_2 × erosion × land cover synthesis
Deep dive	6	Multi-tool environmental profiling
Interpretation	7	Reasoning over analysis results
Error handling	6	Non-existent entities, invalid indicators
Habitat analysis	7	BigEarthNet V2 land cover (Portugal)
Language	6	Galician, Basque inputs
Municipality	4	Municipality-level analysis, disambiguation
Memory (multi-turn)	6	Follow-up questions, context retention
Spatial reasoning	4	Coastal, meseta, community knowledge
Multi-muni ranking	3	Multi-municipality comparisons
Threshold	3	Numeric threshold queries
Temporal change	1	Cross-country temporal context
Error recovery	3	Graceful fallback when data is unavailable
Ranking	2	Top-N and bottom-N province queries
Comparison	2	Close-value and directional comparisons
Single analysis	2	Basic single-province analysis
Province aggregation	2	Aggregate statistics across provinces

Each task specifies: a natural language query (optionally with multi-turn history), expected tool calls, must-contain/must-not-contain strings, maximum rounds, cost budget, and domain-expert ground truth. Error handling tasks are deliberately unsolvable—requesting analysis on non-existent municipalities, fabricating statistics from display-only layers, or citing indicators that do not exist—testing the agent's ability to decline gracefully rather than hallucinate.

The benchmark covers three environmental indicators across two countries:

- **CO_2 absorption suitability** (categorical, Spain): legislative pre-screening encoding five MITECO spatial criteria [9], producing three classes (Not Eligible, Eligible with Conditions, Eligible). Served as a Cloud Optimized GeoTIFF (COG).
- **Gully erosion probability** (continuous, Europe): machine learning prediction from LUCAS 2022 soil survey (0–100%). Served as a COG.
- **BigEarthNet V2 land cover** (categorical, Portugal): 7-class land cover distribution derived from 75k+ labeled Sentinel-2 patches [4], aggregated by Portuguese district. Served as pre-computed JSON statistics.

Seven additional layers are available for visual display but not statistical analysis, creating natural hallucination traps.

3.2 Agent Architecture

Figure 1 shows the system architecture. The agent operates in a ReAct-style loop [25]: it receives a query, reasons about tools, executes calls, observes results, and generates a response. The agent has access to twelve tools (see Table 2).

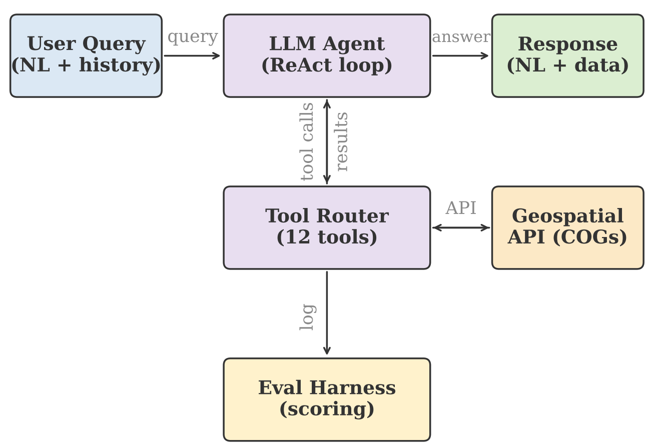

Figure 1: GeoNatureAgent Benchmark system architecture. The LLM agent interacts with a production API serving COG data; the eval harness logs all tool calls and scores responses.

The design choice of structured tool calling over code generation is motivated by empirical evidence: GeoJSON Agents report 85–97% accuracy with function calling [14], while UnivEARTH reports only 33% when agents generate executable code [8].

Table 2: Agent tools. The first eight are domain-specific; the last four are adapted from GeoBenchX [10].

Tool	Purpose
`lookup_province`	Resolve province → boundary geometry
`lookup_municipality`	Resolve municipality (+ province hint)
`analyze_area`	Zonal statistics for indicator in AOI
`analyze_multi_layer`	Multi-indicator analysis in single call
`compare_areas`	Compare two areas on the same indicator
`find_top_n`	Rank provinces by indicator value
`toggle_layer`	Enable/disable map layer visibility
`generate_chart`	Bar/stacked-bar chart from data
`create_buffer`	Buffer geometry by distance (km)
`select_spatial_rel`	Select features by spatial predicate
`get_centroids`	Extract centroid coordinates
`reject_task`	Decline unsolvable tasks

Multi-backend inference client. Fair cross-model comparison requires normalizing heterogeneous API surfaces into a common interface. Our inference client routes models through two backends: a native Anthropic client for Claude and a LiteLLM-based client for Vertex AI Model-as-a-Service (MaaS) endpoints that expose an OpenAI-compatible API. The MaaS backend maps publisher-prefixed model identifiers (e.g., `meta/llama4-...`) to Vertex AI endpoints and normalizes response formats—including tool-call representations—into the shared schema consumed by the eval harness. Notably, Llama 4 models emit tool calls as Python-style function invocations (`<|python_start|>func(k=v)<|python_end|>`) rather than JSON; a dedicated AST-based parser extracts these into structured tool calls so that Llama 4 Scout's responses are scored on the same footing as the JSON-native models. Each inference call uses a layered timeout architecture combining daemon-thread hard timeouts with exponential-backoff retry on rate-limit errors, enabling unattended execution of the full case-seed grid.

3.3 Evaluation Protocol

Each task is scored by eight check types (see Table 3 and Figure 2). A task *passes* only when **all** applicable checks pass (strict binary). We additionally compute partial-credit metrics for finer-grained analysis.

Continuous metrics. Beyond binary pass/fail, we report: *check score*—fraction of individual checks passed per case (partial credit); *keyword coverage*—fraction of `must_contain` terms found; *tool F1*—set-based F1 between expected and actual tools; and *cost utilisation*—ratio of actual to budgeted cost.

Error classification. When a task fails, the first failing check (in priority order: tool_missing → chart_missing → wrong_data → rounds_exceeded → keyword_missing → forbidden_keyword) determines the error category. This priority reflects diagnostic value: tool selection failures are the most informative for understanding agent reasoning. Cases that would only fail `max_cost_usd` are not counted as failures for capability scoring; their cost is instead reported on the cost–accuracy axis (Section 4.4).

Table 3: Scoring checks. Each task specifies which checks apply; binary capability pass requires every applicable check except `max_cost_usd` to pass (Paragraph "Capability and cost as orthogonal axes").

Check	Pass condition
`expected_tools`	Every expected tool was called (recall = 1.0). Extra tools do not cause failure.[a]
`expected_actions`	Every expected UI action was generated (recall = 1.0).
`must_contain`	Each required keyword appears in the answer (case-insensitive substring).
`must_not_contain`	Forbidden keyword does not appear in the answer.
`numeric_accuracy`	For each ground-truth entry, the label is found in the answer and the nearest percentage value is within tolerance.[b]
`chart_generated`	At least one chart URL was produced (only checked when `generate_chart` is an expected tool).
`max_rounds`	Agent loop iterations ≤ budget.
`max_cost_usd`[c]	Estimated token cost ≤ per-case budget. *Reported but not gated* on capability pass/fail.

[a]Tool F1 (precision, recall, F1 on unique tool sets) is computed as a continuous metric but is not used as a gate. [b]A 120-character window after the label is searched for the first `float%` pattern; tolerance defaults to ±2 pp. [c]Per-case budgets default to \$0.10 in the v5 task set. The per-case cost is logged and used in cost–accuracy analysis but is excluded from the binary capability score; see "Capability and cost as orthogonal axes" below.

Design rationale. Tool and action checks use *recall* as the gate rather than F1. An agent that calls all expected tools plus an exploratory extra tool should not be penalised—the key question is whether it performed the required operations. F1 is tracked as a separate metric for efficiency analysis.

Capability and cost as orthogonal axes. Per-case cost budgets are deployment thresholds, not measures of reasoning ability: a model that pays more in tokens has not necessarily reasoned less correctly. Conflating the two into a single binary score forces the reader to interpret a model's accuracy through the lens of one specific, unjustified price threshold. We therefore decouple them: the binary leaderboard scores capability only (does the agent solve the case, evidenced by tool selection, keywords, and numeric accuracy), and per-case cost is reported alongside as a Pareto axis (Section 4.4). This separation follows HELM [3], GAIA [16], and other modern agent benchmarks that report cost and latency as orthogonal dimensions rather than gates. Aggregated cost remains visible in the leaderboard table; a reader who cares about a specific per-case budget can recover the cost-gated number from the supplementary per-case CSV (column `passed_with_cost_gate`).

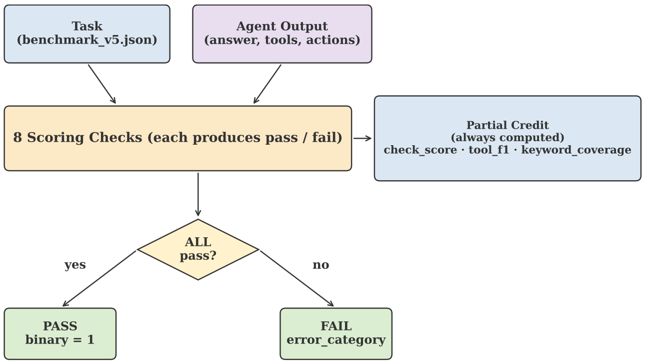

Figure 2: Scoring pipeline. Each case is evaluated by up to eight checks; binary pass requires all to pass. Partial-credit metrics (check score, tool F1, keyword coverage) are always computed.

4 Experiments and Results

4.1 Experimental Setup

We evaluate seven LLMs on GeoNatureAgent Benchmark v5 (93 tasks, 18 categories) across Vertex AI Model-as-a-Service and the Anthropic Messages API (Table 4). All experiments use the v3 system prompt (a zero-shot ReAct-style instruction; the full prompt is committed to the repository at `api/agent/prompts/v3.md`), `max_turns=10`, `max_tokens=4096`, and `temperature=1.0`. We use temperature-1.0 to expose run-to-run variance and stress-test tool-use reliability under realistic sampling noise rather than the deterministic temperature-0 regime that overstates production stability. The prompt is held constant across all models to isolate capability from prompt-engineering effort; chain-of-thought, few-shot, and self-reflection variants are distinct research questions and are left for future work. Experiments are run sequentially inside a single Cloud Run Job to ensure identical infrastructure conditions.

Table 4: Models evaluated. Six via Vertex AI; Claude via the Anthropic Messages API.

Model	Provider	Params	Access
DeepSeek V3.2	DeepSeek AI	671B MoE	Vertex MaaS
GLM-5	Zhipu AI	—	Vertex MaaS
Gemini 2.5 Pro	Google	—	Vertex native
Claude Sonnet 4	Anthropic	—	Anthropic API
Qwen3-235B	Alibaba	235B MoE	Vertex MaaS
GPT-OSS-120B	OpenAI	120B	Vertex MaaS
Llama 4 Scout	Meta	109B MoE	Vertex MaaS

4.2 Aggregation Protocol

To control for run-to-run variance under temperature-1.0 sampling, every model is evaluated on the full v5 case set (93 tasks) under three random seeds, $\{42, 1337, 2024\}$, where the underlying inference API exposes a `seed` parameter (Vertex MaaS endpoints). The Anthropic Messages API does not implement deterministic seeding, so Claude Sonnet 4 is evaluated under two independent temperature-1.0 samples drawn under the same configuration; this yields a valid run-to-run variance estimate but is not bit-for-bit replayable. Reported accuracies are the per-seed mean with one standard deviation, and per-category accuracies are the unweighted mean across the available seeds. Each Cloud Run output directory contains a `results.jsonl` where every row carries its own seed, `case_id`, and `git_commit`, so any leaderboard cell can be traced back to a Cloud Build log line.

The mapping from each model's leaderboard entry to its specific Cloud Run output directory is committed in `paper/final_results/sources.yaml`. A single deterministic script (`scripts/compile_final_results.py`) reads this manifest, pulls each `results.json` from GCS, and emits the per-model leaderboard, per-category breakdown, and per-case matrix as CSV artifacts that are themselves committed to the repository. These CSVs are the single source of truth for every number reported in this paper; a reviewer can regenerate them from the public bucket with one command and diff-check against the committed copies.

4.3 Overall Results

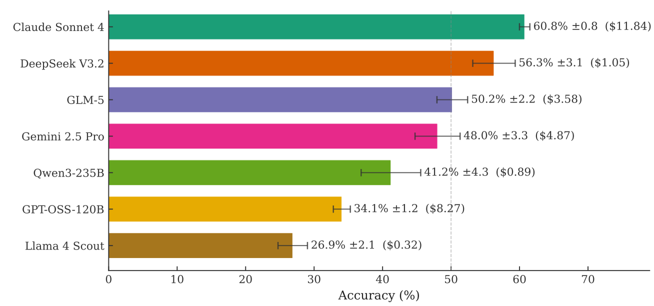

Figure 3: GeoNatureAgent Benchmark v5 leaderboard. Accuracy (%) with total cost in parentheses. Dashed line at 50%.

Table 5: Capability leaderboard: seven models on 93 GeoNatureAgent Benchmark v5 tasks. Accuracy is the per-seed mean with one standard deviation, decoupled from cost (Section 3.3). Cost is the mean total cost of one seed run (93 cases); $/case is the per-call mean.

#	Model	Acc.	Cost	$/case
1	Claude Sonnet 4†	60.8% ± 0.8	$11.84	.127
2	DeepSeek V3.2	56.3% ± 3.1	$1.05	.011
3	GLM-5	50.2% ± 2.2	$3.58	.038
4	Gemini 2.5 Pro	48.0% ± 3.3	$4.87	.052
5	Qwen3-235B	41.2% ± 4.3	$0.89	.010
6	GPT-OSS-120B	34.1% ± 1.2	$8.27	.089
7	Llama 4 Scout	26.9% ± 2.1	$0.32	.003

†Claude Sonnet 4 is reported from two temperature-1.0 samples (the Anthropic Messages API does not implement deterministic seeding); the other six models are evaluated under three seeds. The variance estimate for Claude is therefore based on fewer observations and should be interpreted as an indicative measure rather than a precise one.

Table 5 and Figure 3 summarize the main results aggregated across seeds. **Claude Sonnet 4** leads on capability at 60.8% ± 0.8%, with **DeepSeek V3.2** a close second at 56.3%±3.1%. **GLM-5** (50.2%± 2.2%) and **Gemini 2.5 Pro** (48.0% ± 3.3%) form the second tier. **Qwen3-235B** (41.2% ± 4.3%), **GPT-OSS-120B** (34.1% ± 1.2%), and **Llama 4 Scout** (26.9% ± 2.1%) trail. Only two models exceed 55%, and none reaches 65%, confirming that environmental geospatial tool orchestration remains a significant challenge even for the strongest frontier models. Inter-seed standard deviations are tight (0.8–4.3 percentage points), so the inter-model ordering is robust to sampling noise.

Three findings stand out. First, **the top of the leaderboard is bimodal but close**: Claude Sonnet 4 and DeepSeek V3.2 are separated by only 4.5 percentage points (Welch's $t = 2.40$, $df \approx 2$, $p \approx 0.13$)—within statistical noise at our sample size—while both lead the third-place GLM-5 by more than 6 pp ($t \geq 2.76$). The capability gap between the top two and the rest of the cohort is robust: Claude Sonnet 4 vs. Llama 4 Scout yields $t = 25.0$, and every adjacent pair below the top two is significant at $p < 0.1$ except GLM-5 vs. Gemini 2.5 Pro, which are statistically indistinguishable. Second, **scale does not predict accuracy within the open-weight cluster**: Llama 4 Scout (109B MoE active parameters) trails Qwen3-235B, which trails DeepSeek V3.2 (671B MoE), but GPT-OSS-120B is outperformed by the 109B-parameter Scout on partial-credit metrics, indicating that parameter count alone is insufficient to predict tool-orchestration competence. Third, **closed-source frontier models do not categorically dominate open-weight peers**: Claude leads on capability but only marginally above DeepSeek; Gemini 2.5 Pro sits in third position, below both open-weight contenders. As we show in Section 4.4, this pattern is reinforced on the cost axis, where the entire cost-accuracy Pareto frontier is occupied by open-weight models.

Partial credit tells a different story. Even failing cases often demonstrate partial competence: correct tool selection but missing a keyword, or correct keywords but an extra unnecessary tool call. This gap between binary accuracy and partial-credit scores (Figure 4) suggests that many failures are "near-misses" addressable through prompt engineering.

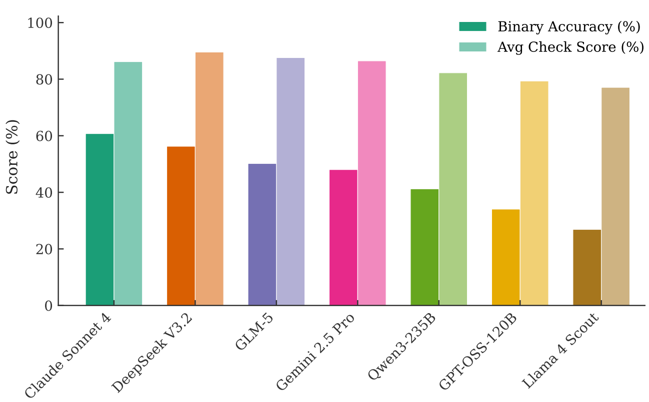

Figure 4: Binary accuracy vs partial-credit check score. The gap indicates many "near-miss" failures.

4.4 Cost-Accuracy Analysis

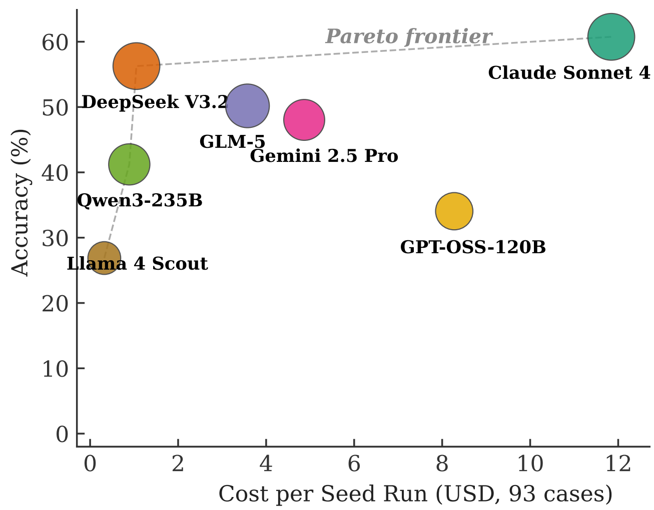

Figure 5: Cost-accuracy trade-off. Bubble size is proportional to total tokens; cost is the mean per-seed run cost (93 cases). The Pareto frontier (dashed line) runs Scout → Qwen3-235B → DeepSeek V3.2 → Claude Sonnet 4: three of the four frontier models are open-weight.

Cost per case varies by more than an order of magnitude across the cohort, from \$0.003 for Llama 4 Scout to \$0.127 for Claude Sonnet 4 (see Figure 5). The cost-accuracy Pareto frontier comprises four models: **Llama 4 Scout** (cheapest, 26.9% capability), **Qwen3-235B** (41.2% at \$0.010/case), **DeepSeek V3.2** (56.3% at \$0.011/case), and **Claude Sonnet 4** (highest capability at 60.8%, \$0.127/case). Three of these four frontier models are open-weight; closed-source Gemini 2.5 Pro and the open-weight GPT-OSS-120B are dominated (every Pareto-frontier point matches or exceeds their accuracy at lower cost). The DeepSeek V3.2 entry is particularly notable: it delivers 93% of Claude's capability (56.3% vs. 60.8%) at 11.6× lower per-case cost, making it the strongest single-model recommendation for cost-sensitive deployments. Models priced between DeepSeek and Claude—GLM-5, Gemini 2.5 Pro, and GPT-OSS-120B—neither match Claude's accuracy nor undercut DeepSeek's price; they are interior to the frontier in the cost-accuracy plane.

Table 6: Projected per-query inference cost at 1,000 queries/day. Daily and monthly data extrapolate the per-case cost reported in Table 5. Sorted by capability accuracy.

Model	Accuracy	Daily	Monthly
Claude Sonnet 4	60.8%	\$127	\$3,810
DeepSeek V3.2	56.3%	\$11	\$330
GLM-5	50.2%	\$38	\$1,140
Gemini 2.5 Pro	48.0%	\$52	\$1,560
Qwen3-235B	41.2%	\$10	\$300
GPT-OSS-120B	34.1%	\$89	\$2,670
Llama 4 Scout	26.9%	\$3	\$90

Table 6 projects per-query costs to a production deployment serving 1,000 queries/day. The capability leader (Claude Sonnet 4, 60.8%) costs $127/day, while the second-place DeepSeek V3.2 (56.3%) costs $11/day—a 4.5-percentage-point capability gap for an 11.6× price difference. For most production deployments, the right operating point sits on the open-weight Pareto frontier; the rare scenarios that justify the premium price are those where the absolute capability ceiling matters more than throughput economics. These dynamics align with prior cost-efficient findings such as Geo-OLM [21], which shows that small specialized models can match frontier model performance at a fraction of frontier cost; here, the same pattern emerges among general-purpose models at the larger scale required for environmental analysis.

Figure 6 further illustrates this relationship, showing that token consumption does not predict accuracy—models that generate more tokens do not necessarily achieve higher scores.

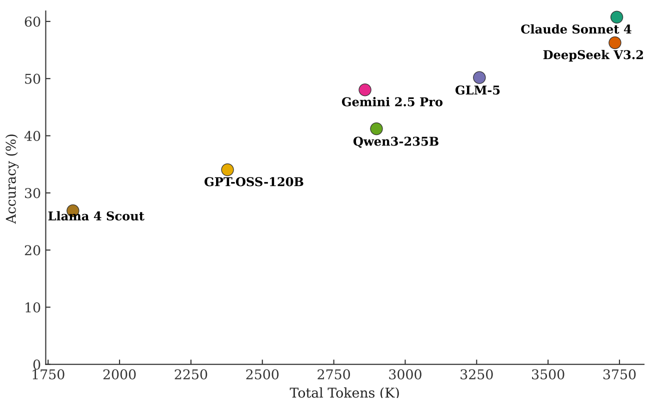

Figure 6: Token consumption vs. accuracy. Token volume is a poor predictor of performance.

4.5 Failure Analysis

4.5.1 Universally Hard Tasks. One case—close-value comparison—defeats every model in every seed, and two further categories sit in the 0–25% range (Figure 7), revealing systematic limitations:

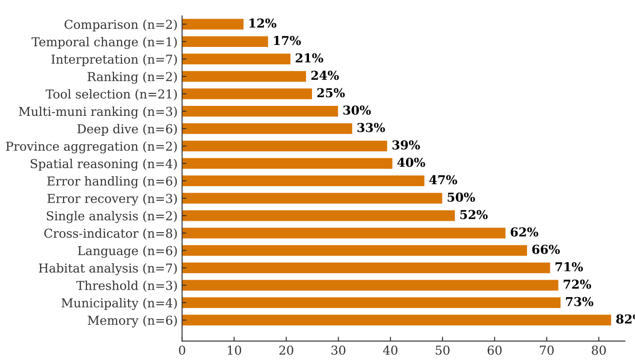

Figure 7: Category difficulty—mean accuracy across all seven models, sorted by difficulty.

Close-value comparison (case `V5_21`, 0/20 trials across 7 models × 2–3 seeds): the close-value sub-case (Murcia 68.5% vs. Córdoba 68.4%) leads every model in every seed to fabricate a directional difference rather than reporting near-identical values. We treat this as a strong qualitative falsification of the hypothesis that any model in our cohort returns "effectively equal" as an answer; the failure mode is consistent: agents call the correct `compare_areas` tool, receive the correct numeric output, and then state in natural language that one province is higher. The *directional* sub-case (`V5_22`: Asturias-north vs. Granada-south erosion, where the difference is large and obvious) tells a different story: DeepSeek V3.2 passes 2/3 seeds and GPT-OSS-120B passes 3/3, while the other five models still fail. The capability gap is therefore specifically about distinguishing "different" from "effectively equal" under small numeric gaps, not about directional reasoning in general. **Ranking** (0–33% across models): bottom-N and top-N province queries failed even on models that aced single-province lookups, with errors split between calling the wrong tool (`analyze_area` instead of `find_top_n`) and confabulating rankings from training data. **Single analysis** (0–100%): trivially-passable for some models but a 0/2 floor for Claude and Llama 4 Scout. **Province aggregation** (0–75%): wide variance, with Claude and GLM-5 above 50% but DeepSeek and Qwen3 below 20%.

Conversely, the easiest categories are **habitat analysis** (Claude 100%, GLM-5 81%, mean 60%), **memory/multi-turn** (mean 79%), and **language** (mean 51% across the five cost-gate-free models; Claude's 0% on this category is a cost-budget artefact, Section 4.4). Models that pay the cost-budget can sustain multi-turn context and parse Galician/Basque inputs but cannot reliably perform the multi-step numeric reasoning that ranking and comparison demand.

4.5.2 Model-Specific Patterns. **DeepSeek V3.2** (56.3%): leader, with the most balanced profile—above 70% on cross-indicator synthesis, memory, multi-municipality ranking, single analysis, spatial reasoning, threshold, and municipality categories. Notably the only model in the top three to break 30% on the universally hard comparison category (33%, vs 0% for all others). **GLM-5** (50.2%): strong on cross-indicator (88%), habitat (81%), memory (83%), municipality (92%), and threshold (78%); weaker on tool selection (27%) and a 0% floor on comparison and ranking. **Gemini 2.5 Pro** (48.0%): the highest-accuracy closed-source model in the cohort, with habitat (86%), memory (78%), and error handling (72%) among its strengths; notably weak on ranking and temporal change (both 0%). **Claude Sonnet 4** (60.8%): the capability leader, with the most uniformly strong per-category profile in the cohort. Claude reaches 100% on habitat analysis, memory/multi-turn, and municipality; 94% on cross-indicator synthesis (the highest in the cohort); 83% on threshold; and 75% on language, spatial reasoning, and province aggregation. Its principal weakness is the universally hard comparison category (0%, shared with five other models—only DeepSeek breaks zero) and tool selection (33%, tied with DeepSeek but well below its other categories). Per-token cost is high ($0.127/case), placing Claude on the upper-right of the Pareto frontier: the most capable single point, but at 11.6× the price of the next-best Pareto point (DeepSeek). **Qwen3-235B** (41.2%): strongest on threshold (100%), habitat (76%), memory (83%), and deep dive (56%); weakest on tool selection (17%) and province aggregation (0%). **GPT-OSS-120B**

(30.1%): moderate across categories with no standout strengths; threshold (89%) is its only category above 70%. **Llama 4 Scout** (26.9%): trails the cohort, with successes concentrated in language (78%), memory (72%), and error recovery (67%) but a near-floor on tool selection (6%) and cross-indicator synthesis (29%).

4.6 Category-Level Analysis

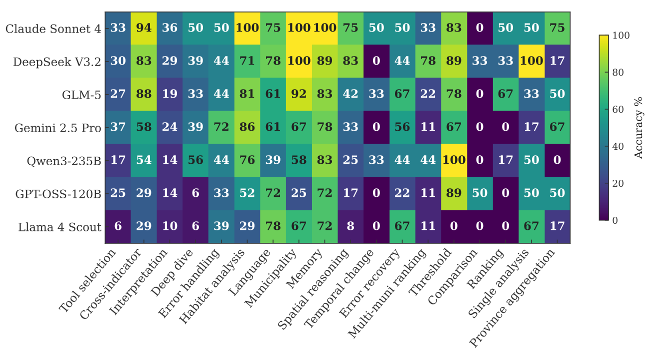

Figure 8: Accuracy by category across the seven models. Comparison achieves 0% across every model; the largest category (tool selection, 21 tasks) shows large inter-model variance.

Figure 8 shows mean accuracy by category across all seven models. Three structural findings emerge:

(1) Close-value comparison is universally unsolved. The close-value sub-case (V5_21) yields 0/20 across all model-seed trials; the directional sub-case (V5_22) is partly solvable (DeepSeek 67%, GPT-OSS-120B 100%; all other models 0%). **(2) Tool selection variance is the dominant cross-model spread.** The 21-case tool selection category is the largest and the most discriminative: Claude (33%), DeepSeek (30%), GLM-5 (27%), GPT-OSS-120B (25%) cluster within an 8-point band, while Qwen3-235B (17%) and Scout (6%) trail. **(3) Model strengths are domain-specialised rather than uniform.** No model dominates every category. Claude leads on habitat analysis (100%), memory (100%), municipality (100%), cross-indicator synthesis (94%), and ties on a further three categories. DeepSeek leads on single analysis (100%), spatial reasoning (83%), multi-municipality ranking (78%), and is the unique non-zero on close-value comparison sub-cases (33%). GLM-5 leads on threshold (78%) and is the only mid-pack model to clear 65% on cross-indicator (88%). Gemini leads on error handling (72%). Qwen3 ties Claude on threshold (100% on three cases). A practitioner optimising deployment for a specific category should consult the per-category breakdown rather than the headline accuracy alone—no single model is uniformly best, and the right choice depends on which task patterns dominate the target workload.

5 Discussion

5.1 Comparison with Prior Benchmarks

GeoNatureAgent Benchmark's 60.8% best-model accuracy is notably lower than GeoBenchX and GeoJSON Agents (85–97%). Table 7 summarizes the key differences.

Table 7: Comparison of structured tool-calling geospatial benchmarks.

	GeoBenchX	GeoNatureAgent Benchmark
Tasks	202	93
Categories	5	18
Tools	24 primitives	12 domain ops
Tool abstraction	Low-level	High-level
Data source	Local files	Cloud API (COG + JSON)
Domain	General GIS	Environmental
Countries	—	Spain + Portugal
Indicators	—	3
Multi-turn	No	Yes
Best accuracy	85–90%	60.8%
Evaluation	LLM-as-judge	Metric-based

Four factors explain the accuracy gap: (1) real API interaction vs. local file operations; (2) environmental domain complexity (geographic knowledge, data limitations, display-only layers); (3) multi-turn conversation history; and (4) strict binary evaluation. Our results align with harder benchmarks: MapEval (<67%) [5], SpatiaLab (55%) [24], and UnivEARTH (33%) [8].

The GeoBenchX comparison is particularly instructive because both benchmarks share the structured tool-calling paradigm. The roughly 30-percentage-point accuracy drop from general GIS tasks to environmental analysis suggests that general-purpose geospatial benchmarks substantially overestimate agent capability for domain-specific workflows. Environmental tasks require geographic knowledge (e.g., which provinces have erosion data), awareness of data limitations (display-only layers that cannot be analyzed), multi-step reasoning across indicators, and temporal awareness—challenges absent from generic filter-merge-visualize pipelines. This motivates domain-specific benchmarks as a necessary complement to general-purpose evaluations.

5.2 Structured Tool Calling vs. Code Generation

Our 60.8% best-model accuracy on real API tool calls compares favorably with UnivEARTH's 33% on code generation [8]. While not directly comparable, the structured approach constrains the output space, eliminating the 58% code execution failure rate that UnivEARTH reports. This provides further evidence that production environmental systems should prefer structured APIs over free-form code generation.

5.3 Cross-Dataset Evaluation: BigEarthNet V2

GeoNatureAgent Benchmark is designed to be domain- and geography-agnostic. To demonstrate extensibility beyond Spanish CO_2 and erosion indicators, we incorporate BigEarthNet V2 [4] as a third data domain covering a different country. BigEarthNet V2 provides 549k labeled Sentinel-2 patches with CLC2018 19-class labels; we aggregate the 75k+ Portuguese patches across 9 mainland districts into a 7-class land cover distribution (Coniferous Forest, Broadleaf

Forest, Shrubland, Grassland, Sparse Vegetation, Water, and Urban) queryable via the same `analyze_area` tool used for Spanish indicators.

This integration tests whether agents generalize from Spanish environmental data to a different country and data domain. The 20 BigEarthNet benchmark cases (Table 8) span seven task patterns: single-district breakdown, cross-indicator synthesis (Portuguese land cover with Spanish CO_2/erosion), multi-turn recall, chart generation, legend and layer discovery, temporal context, and two-district comparison. Unlike the CO_2 and erosion indicators, where ground truth depends on live raster geometry, BigEarthNet cases have fixed numeric ground truth derived from pre-computed district-level statistics, enabling deterministic scoring.

Table 8: BigEarthNet V2 benchmark cases by task pattern. All cases use Portuguese districts with pre-computed ground truth.

Task Pattern	Cases
Single-district land cover breakdown	3
Cross-indicator (PT land cover + ES)	3
Multi-turn recall from session history	2
Chart generation	2
Legend / list layers discovery	2
Deep-dive multi-indicator profile	4
Two-district comparison	4

While Table 8 lists the 20 purpose-built BigEarthNet cases, Portuguese land cover data also appears in 4 additional cases across other categories (tool selection, interpretation, and error recovery), yielding 24 BigEarthNet-related cases in total. Aggregated across seeds, **Claude Sonnet 4** achieves the highest mean accuracy on this subset (100% on the dedicated habitat-analysis cases and 94% on cross-indicator cases that combine Portuguese land cover with Spanish indicators), followed by **GLM-5** (habitat 81%, cross-indicator 88%) and **DeepSeek V3.2** (habitat 71%, cross-indicator 83%). Claude's standout performance on the BigEarthNet cases—consistent with its top position on the overall capability leaderboard—suggests that its instruction-following and JSON-handling reliability are particularly well-suited to the pre-computed district-level indicator workflow. The BigEarthNet subset also illustrates a broader point: adding a third indicator and a second country required no engine changes, validating the framework's domain-and-geography-agnostic claim.

The inclusion of BigEarthNet V2 as a queryable layer rather than merely a validation dataset transforms the benchmark's geographic scope from Spain-only to cross-country (Spain + Portugal), and its indicator count from two to three. This validates the claim that the framework generalizes to new data sources: adding a new indicator required only a JSON statistics file, a prompt update, and new benchmark cases———no changes to the scoring engine or evaluation protocol.

5.4 Limitations

Evaluation strictness: binary pass/fail may understate capability (Section 3.3 describes the partial-credit metrics that address this). **Architecture scope**: every system evaluated is a single-agent ReAct loop; multi-agent decompositions (Planner-Worker, Squad architectures) are known to deliver large gains on adjacent benchmarks [12, 14] and would be a natural extension.

6 Conclusion

We introduce GeoNatureAgent Benchmark, a benchmark for evaluating AI agents on environmental geospatial analysis through structured tool calling against an open, self-hostable API. The benchmark spans three indicators across Spain and Portugal, demonstrating domain and geographic extensibility. Evaluating seven LLMs on 93 tasks across 18 categories with twelve tools, under three temperature-1.0 seeds per model and reporting capability and per-case cost as orthogonal axes:

(1) **Environmental geospatial tool orchestration remains open.** The capability leader (Claude Sonnet 4) achieves 60.8%± 0.8%, well below the 85–97% reported on general GIS benchmarks; only two models exceed 55% and none reaches 65%.
(2) **Open-weight models occupy the cost–accuracy Pareto frontier.** Three of the four frontier models (Llama 4 Scout, Qwen3-235B, DeepSeek V3.2) are open-weight; closed-source Gemini 2.5 Pro and the larger open-weight GPT-OSS-120B are dominated.
(3) **The capability–cost trade-off is steep.** DeepSeek V3.2 delivers 93% of Claude's capability (56.3% vs. 60.8%) at 11.6× lower per-case cost, making it the strongest single-model recommendation for cost-sensitive production deployment.
(4) **Systematic failures persist.** Every model scores 0% on close-value comparison; tool-selection variance remains the dominant inter-model spread; parameter scale alone does not predict capability.

Future work: expanding to additional countries and indicators (NDVI, precipitation, and biodiversity indices); prompt ablation (zero-shot, CoT, few-shot); multi-agent architecture comparison following the Planner-Worker paradigm of GeoJSON Agents [14]; and dedicated mitigation strategies for the universally-unsolved comparison category.

GeoNatureAgent Benchmark, evaluation code, and the full MLOps pipeline are publicly available.[2]

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References

[1] Temitope Akinboyewa, Zhenlong Li, Huan Ning, and M. Naser Lessani. 2025. GIS Copilot: Towards an Autonomous GIS Agent for Spatial Analysis. *International Journal of Digital Earth* 18, 1 (2025), 2497489. doi:10.1080/17538947.2025.2497489

[2] Christian Michael Arnold, Andrew Alini, Jonathan Wang, Pieter M. Feenstra, Conner Arnold, Jan DeWitt, Natalie C. Ritsema, Jung Hyun Yae, Gary C. Borchardt, Boris Katz, Andrei Barbu, and Brian Cheung. 2026. MapQA: A Map-Question-Answering Benchmark for Visual Language Model Reasoning. https://openreview.net/forum?id=dOISCbmkmG

[2] https://github.com/gabrielireland/GeoNatureAgent_Benchmark

[3] Rishi Bommasani, Percy Liang, and Tony Lee. 2023. Holistic Evaluation of Language Models. *Annals of the New York Academy of Sciences* 1525, 1 (2023), 140–146. arXiv:https://nyaspubs.onlinelibrary.wiley.com/doi/pdf/10.1111/nyas.15007 doi:10.1111/nyas.15007

[4] Kai Norman Clasen, Leonard Hackel, Tom Burgert, Gencer Sumbul, Begüm Demir, and Volker Markl. 2025. reBen: Refined BigEarthNet Dataset for Remote Sensing Image Analysis. In *IGARSS 2025 – 2025 IEEE International Geoscience and Remote Sensing Symposium*. 1264–1268. doi:10.1109/IGARSS55030.2025.11242834

[5] Mahir Labib Dihan, Md Tanvir Hassan, Md Tanvir Parvez, Md Hasebul Hasan, Md Almash Alam, Muhammad Aamir Cheema, Mohammed Eunus Ali, and Md Rizwan Parvez. 2025. MapEval: A Map-Based Evaluation of Geo-Spatial Reasoning in Foundation Models. In *Proceedings of the 42nd International Conference on Machine Learning* (Vancouver, Canada) *(ICML'25)*. JMLR.org, Article 526, 40 pages.

[6] Google Research. 2025. Google Earth AI: Unlocking Geospatial Insights with Foundation Models and Cross-Modal Reasoning. Google Research Blog. https://research.google/blog/google-earth-ai-unlocking-geospatial-insights-with-foundation-models-and-cross-modal-reasoning/

[7] Yu Huang, Liang Guo, Wanqian Guo, Zhe Tao, Yang Lv, Zhihao Sun, and Dongfang Zhao. 2024. EnviroExam: Benchmarking Environmental Science Knowledge of Large Language Models. arXiv:2405.11265 [cs.CL] doi:10.48550/arXiv.2405.11265

[8] Chia Hsiang Kao, Wenting Zhao, Shreelekha Revankar, Samuel Speas, Snehal Bhagat, Rajeev Datta, Cheng Perng Phoo, Utkarsh Mall, Carl Vondrick, Kavita Bala, and Bharath Hariharan. 2025. Towards LLM Agents for Earth Observation. arXiv:2504.12110 [cs.AI] doi:10.48550/arXiv.2504.12110

[9] Kingdom of Spain. 2025. Real Decreto 214/2025 on the Regulation of the Carbon Footprint Register. Boletín Oficial del Estado. https://www.boe.es/buscar/doc.php?id=BOE-A-2025-7439

[10] Varvara Krechetova and Denis Kochedykov. 2025. GeoBenchX: Benchmarking LLMs in Agent Solving Multistep Geospatial Tasks. In *Proceedings of the 1st ACM SIGSPATIAL International Workshop on Generative and Agentic AI for Multi-Modality Space-Time Intelligence (GeoGenAgent '25)*. Association for Computing Machinery, New York, NY, USA, 27–35. doi:10.1145/3764915.3770721

[11] Alexandre Lacoste, Nils Lehmann, Pau Rodriguez, Evan David Sherwin, Hannah Kerner, Björn Lütjens, Jeremy Irvin, David Dao, Hamed Alemohammad, Alexandre Drouin, Mehmet Gunturkun, Gabriel Huang, David Vazquez, Dava Newman, Yoshua Bengio, Stefano Ermon, and Xiao Xiang Zhu. 2023. GEO-Bench: Toward Foundation Models for Earth Monitoring. In *Proceedings of the 37th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) *(NIPS '23)*. Curran Associates Inc., Red Hook, NY, USA, Article 2223, 14 pages.

[12] Chaehong Lee, Varatheepan Paramanayakam, Andreas Karatzas, Yanan Jian, Michael Fore, Heming Liao, Fuxun Yu, Ruopu Li, Iraklis Anagnostopoulos, and Dimitrios Stamoulis. 2025. Multi-Agent Geospatial Copilots for Remote Sensing Workflows. In *IGARSS 2025 – 2025 IEEE International Geoscience and Remote Sensing Symposium*. 1084–1089. doi:10.1109/IGARSS55030.2025.11243915

[13] Zhenlong Li and Huan Ning. 2023. Autonomous GIS: The Next-Generation AI-Powered GIS. *International Journal of Digital Earth* 16, 2 (2023), 4668–4686. doi:10.1080/17538947.2023.2278895

[14] Qianqian Luo, Qingming Lin, Liuchang Xu, Sensen Wu, Ruichen Mao, Chao Wang, Hailin Feng, Bo Huang, and Zhenhong Du. 2026. GeoJSON Agents: A Multi-Agent LLM Architecture for Geospatial Analysis—Function Calling vs. Code Generation. *Big Earth Data* 0, 0 (2026), 1–55. doi:10.1080/20964471.2026.2615511

[15] Valerio Marsocci, Yuru Jia, Georges Le Bellier, David Kerekes, Liang Zeng, Sebastian Hafner, Sebastian Gerard, Eric Brune, Ritu Yadav, Ali Shibli, Heng Fang, Yifang Ban, Maarten Vergauwen, Nicolas Audebert, and Andrea Nascetti. 2025. PANGAEA: A Global and Inclusive Benchmark for Geospatial Foundation Models. arXiv:2412.04204 [cs.CV] doi:10.48550/arXiv.2412.04204

[16] Grégoire Mialon, Clémentine Fourrier, Thomas Wolf, Yann LeCun, and Thomas Scialom. 2024. GAIA: a benchmark for General AI Assistants. In *The Twelfth International Conference on Learning Representations*. https://openreview.net/forum?id=fibxvahvs3

[17] Yu-Erh Pan and Ayesha Siddika Nipu. 2025. LLM-Enhanced Air Quality Monitoring Interface via Model Context Protocol. In *2025 7th International Symposium on Advanced Electrical and Communication Technologies (ISAECT)*. IEEE, 1–6. doi:10.1109/ISAECT68904.2025.11318775

[18] Planet Labs and Anthropic. 2025. Planet and Anthropic Partner to Use Claude's Advanced AI Capabilities to Turn Geospatial Satellite Imagery into Actionable Insights. BusinessWire. https://www.businesswire.com/news/home/20250306606139/en/Planet-and-Anthropic-Partner-to-Use-Claudes-Advanced-AI-Capabilities-to-Turn-Geospatial-Satellite-Imagery-into-Actionable-Insights

[19] Akashah Shabbir, Muhammad Akhtar Munir, Akshay Dudhane, Muhammad Umer Sheikh, Muhammad Haris Khan, Paolo Fraccaro, Juan Bernabe Moreno, Fahad Shahbaz Khan, and Salman Khan. 2026. ThinkGeo: Evaluating Tool-Augmented Agents for Remote Sensing Tasks. arXiv:2505.23752 [cs.CV] doi:10.48550/arXiv.2505.23752

[20] Naomi Simumba, Nils Lehmann, Paolo Fraccaro, Hamed Alemohammad, Geeth De Mel, Salman Khan, Manil Maskey, Nicolas Longepe, Xiao Xiang Zhu, Hannah Kerner, Juan Bernabe-Moreno, and Alexandre Lacoste. 2026. GEO-Bench-2: From Performance to Capability, Rethinking Evaluation in Geospatial AI. arXiv:2511.15658 [cs.CV] doi:10.48550/arXiv.2511.15658

[21] Dimitrios Stamoulis and Diana Marculescu. 2025. Geo-OLM: Enabling Sustainable Earth Observation Studies with Cost-Efficient Open Language Models and State-Driven Workflows. In *Proceedings of the 2025 ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies (COMPASS '25)*. Association for Computing Machinery, New York, NY, USA, 608–619. doi:10.1145/3715335.3735494

[22] Daniela Szwarcman, Sujit Roy, Paolo Fraccaro, Thorsteinn Elí Gíslason, Benedikt Blumenstiel, Rinki Ghosal, Pedro Henrique de Oliveira, Joao Lucas de Sousa Almeida, Rocco Sedona, Yanghui Kang, Srija Chakraborty, Sizhe Wang, Carlos Gomes, Ankur Kumar, Vishal Gaur, Myscon Truong, Denys Godwin, Sam Khallaghi, Hyunho Lee, Chia-Yu Hsu, Ata Akbari Asanjan, Besart Mujeci, Disha Shidham, Rufai Omowunmi Balogun, Venkatesh Kolluru, Trevor Keenan, Paulo Arevalo, Wenwen Li, Hamed Alemohammad, Pontus Olofsson, Timothy Mayer, Christopher Hain, Robert Kennedy, Bianca Zadrozny, David Bell, Gabriele Cavallaro, Campbell Watson, Manil Maskey, Rahul Ramachandran, and Juan Bernabe Moreno. 2026. Prithvi-EO-2.0: A Versatile Multitemporal Foundation Model for Earth Observation Applications. *IEEE Transactions on Geoscience and Remote Sensing* 64 (2026), 1–20. doi:10.1109/TGRS.2025.3642610

[23] Thinh Hung Truong, Jey Han Lau, and Jianzhong Qi. 2026. GPSBench: Do Large Language Models Understand GPS Coordinates? arXiv:2602.16105 [cs.AI] doi:10.48550/arXiv.2602.16105

[24] Azmine Toushik Wasi, Wahid Faisal, Abdur Rahman, Mahfuz Ahmed Anik, Munem Shahriar, Mohsin Mahmud Topu, Sadia Tasnim Meem, Rahatun Nesa Priti, Sabrina Afroz Mitu, Md. Iqramul Hoque, Shahriyar Zaman Ridoy, Mohammed Eunus Ali, Majd Hawasly, Mohammad Raza, and Md Rizwan Parvez. 2026. SpatiaLab: Can Vision-Language Models Perform Spatial Reasoning in the Wild? arXiv:2602.03916 [cs.CV] doi:10.48550/arXiv.2602.03916

[25] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *Proceedings of the 11th International Conference on Learning Representations, ICLR 2023*. doi:10.48550/arXiv.2210.03629

[26] Yifan Zhang, Jingxuan Li, Zhiyun Wang, Zhengting He, Qingfeng Guan, Jianfeng Lin, and Wenhao Yu. 2025. Geospatial Large Language Model Trained with a Simulated Environment for Generating Tool-Use Chains Autonomously. *International Journal of Applied Earth Observation and Geoinformation* 136 (2025), 104312. doi:10.1016/j.jag.2024.104312

[27] Yuanxin Zhang, Sijie Lin, Yaxin Xiong, Nan Li, Lijin Zhong, Longzhen Ding, and Qing Hu. 2025. Fine-Tuning Large Language Models for Interdisciplinary Environmental Challenges. *Environmental Science and Ecotechnology* 27 (2025), 100608. doi:10.1016/j.ese.2025.100608

[28] Yifan Zhang, Cheng Wei, Zhengting He, and Wenhao Yu. 2024. GeoGPT: An Assistant for Understanding and Processing Geospatial Tasks. *International Journal of Applied Earth Observation and Geoinformation* 131 (2024), 103976. doi:10.1016/j.jag.2024.103976

A Online Resources

Datasets, code and results related to this article can be found at: https://github.com/gabrielireland/GeoNatureAgent_Benchmark.